ECO LEGACY A ENVIRONMENT & HERITAGE LEARNING SYSTEM

A MAIN PROJECT REPORT

Submitted by

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MLM24MCA-2052

To

The APJ Abdul Kalam Technological University
in Partial fulfillment of the requirements for the
award of the degree

of

MASTER OF COMPUTER APPLICATIONS

Under the Guidance of

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DEPARTMENT OF COMPUTER APPLICATIONS

MANGALAM COLLEGE OF ENGINEERING, ETTUMANOOR

(Affiliated to APJ Abdul Kalam Technological University)

APRIL 2026

MANGALAM COLLEGE OF ENGINEERING

Accredited by NAAC& ISO 9001:2000 Certified Institution

DEPARTMENT OF COMPUTER APPLICATIONS

VISION

To become a centre of excellence in computer applications,competent in the global ecosystem with technical knowledge,innovation with a sense of social commitment.

MISSION

- To serve with state of the art education,foster advanced research and cultivate innovation in the field of computer applications.
- To prepare learners with knowledge skills and critical thinking to excel in the technological landscape and contribute positively to society.

Program Educational Objectives

- PEO I :Graduates will possess a solid foundation and in-depth understanding of computer applications and will be equipped to analyze real-world problems, design and create innovative solutions, and effectively manage and maintain these solutions in their professional careers.
- PEO II: Graduates will acquire technological advancements through continued education, lifelong learning and research, thereby making meaningful contributions to the field of computing.
- PEO III: Graduates will cultivate team spirit, leadership, communication skills, ethics, and social values, enabling them to apply their understanding of the societal impacts of computer applications effectively.

Program Specific Outcomes

- **PSO I**: Apply advanced technologies through innovations to enhance the efficiency of design development.
- **PSO II**: Apply the principles of computing to analyze, design and implement sustainable solutions for real world challenges.

MAPPING OF PO-PSO-SDG

1. MAPPING WITH PROGRAM OUTCOMES (POs):

SL.NO	POs ADDRESSED	RELEVANCE TO PROJECT
1	PO1- Foundation Knowledge	Applying Python, Django, and AI transformer models to solve the problem of automated content generation
2	PO2-Problem Analysis	Identifying the gap in regional heritage preservation and designing a gamified system to address student engagement.
3	PO3-Development of Solutions	Designing the "Knowledge Bloom" UI and the multi- role (Admin/Teacher/Student) architecture of the web platform.
4	PO5- Usage of Modern Tools	Using modern frameworks like Django, VS Code, GitHub for version control, and Hugging Face libraries for AI integration.

LIST OF PROGRAM OUTCOMES (POs):

PO1 (Foundation Knowledge): Apply knowledge of mathematics, programming logic and coding fundamentals for solution architecture and problem solving**.**

PO2 (Problem Analysis): Identify, review, formulate and analyses problems for primarily focusing on customer requirements using critical thinking frameworks.

PO3 (Development of Solutions): Design, develop and investigate problems with as an innovative approach for solutions incorporating ESG/SDG goals**.**

PO4 (Modern Tool Usage): Select, adapt and apply modern computational tools such as development of algorithms with an understanding of the limitations including human biases.

PO5 (Individual and Teamwork): Function and communicate effectively as an individual or a team leader in diverse and multidisciplinary groups. Use methodologies such as agile.

PO6 (Project Management and Finance): Use the principles of project management such as scheduling, work breakdown structure and be conversant with the principles of Finance for profitable project management.

PO7 (Ethics): Commit to professional ethics in managing software projects with financial aspects. Learn to use new technologies for cyber security and insulate customers from malware.

PO8 (Life-long learning): Change management skills and the ability to learn, keep up with contemporary technologies and ways of working**.**

2. MAPPING WITH PROGRAM SPECIFIC OUTCOMES (PSOs):

SL.NO	PSOs ADDRESSED	RELEVANCE TO PROJECT
1	PSO 1	project applies advanced AI and web technologies to automate, optimize, and enhance the design and delivery of educational content, making the system more efficient and innovative.
2	PSO 2	Applies computing principles to analyze educational challenges, design an intelligent system, and implement a scalable and sustainable AI-driven learning platform

LIST OF PROGRAM SPECIFIC OUTCOMES (PSOs):

PSO 1: Apply advanced technologies through innovations to enhance the efficiency of design development.

PSO 2: Apply the principles of computing to analyze, design and implement sustainable solutions for real world challenges.

2. MAPPING WITH SUSTAINABLE DEVELOPMENT GOALS (SDGs):

SDG NO	SDGs ADDRESSED	RELEVANCE TO PROJECT
SDG 4	Quality Education	By providing a specialized LMS that uses AI to generate quizzes and lessons, making heritage and environmental education more accessible and engaging.
SDG 9	Industry, Innovation, and Infrastructure	By developing an AI-driven, scalable, and cost-effective digital learning platform that strengthens educational infrastructure, promotes innovation through advanced technologies, and ensures inclusive access to knowledge.
SDG 11	Sustainable Cities and Communities	By digitizing and preserving Kerala's folklore and regional history, the project protects the world's cultural and natural heritage.

SUSTAINABLE DEVLOPMENT GOALS (SDGs):

SDG 1 – No Poverty-End poverty in all its forms everywhere.

SDG 2 – Zero Hunger-End hunger, achieve food security and improved nutrition, and promote sustainable agriculture.

SDG 3 – Good Health and Well-Being-Ensure healthy lives and promote well-being for all at all ages.

SDG 4 – Quality Education-Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.

SDG 5 – Gender Equality-Achieve gender equality and empower all women and girls.

SDG 6 – Clean Water and Sanitation-Ensure availability and sustainable management of water and sanitation for all.

SDG 7 – Affordable and Clean Energy-Ensure access to affordable, reliable, sustainable, and modern energy for all.

SDG 8 – Decent Work and Economic Growth-Promote sustained, inclusive, and sustainable economic growth, full and productive employment, and decent work for all.

SDG 9 – Industry, Innovation, and Infrastructure-Build resilient infrastructure, promote inclusive and sustainable industrialization, and foster innovation.

SDG 10 – Reduced Inequality-Reduce inequality within and among countries.

SDG 11 – Sustainable Cities and Communities-Make cities and human settlements inclusive, safe, resilient, and sustainable.

SDG 12 – Responsible Consumption and Production-Ensure sustainable consumption and production patterns.

SDG 13 – Climate Action-Take urgent action to combat climate change and its impacts.

SDG 14 – Life Below Water-Conserve and sustainably use the oceans, seas, and marine resources.

SDG 15 – Life on Land -Protect, restore, and promote sustainable use of terrestrial ecosystems, manage forests sustainably, combat desertification, halt and reverse land degradation, and halt biodiversity loss.

SDG 16 – Peace, Justice, and Strong Institutions- Promote peaceful and inclusive societies, provide access to justice for all, and build effective, accountable, and inclusive institutions.

MANGALAM COLLEGE OF ENGINEERING, ETTUMANOOR
DEPARTMENT OF COMPUTER APPLICATIONS
APRIL 2026

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DECLARATION

*I hereby certify that the work which is being presented in the project entitled ‘‘**ECO LEGACY A ENVIRONMENT & HERITAGE LEARNING SYSTEM’’** submitted in the **DEPARTMENT OF COMPUTER APPLICATIONS** is an authentic record of my own work carried under the supervision of **Ms. Banu Sumayya S, Assistant professor**. This study has not been submitted to any other institution or university for the award of any other degree. This report has been checked for plagiarism by the college and the similarity index is within permissible limits set by the college.*

Name & Signature of Student

Date:

Place:

MANGALAM COLLEGE OF ENGINEERING, ETTUMANOOR
DEPARTMENT OF COMPUTER APPLICATIONS
APRIL 2026

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CERTIFICATE

Certified that the report entitled ''ECO LEGACY ENVIRONMENT & HERITAGE LEARNING SYSTEM'' submitted by **SHYAM KRISHNAN (MLM24MCA-2052)** to the APJ Abdul Kalam Technological University in partial fulfilment of the requirements for the award of the **DEGREE MASTER OF COMPUTER APPLICATIONS** by **APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY** is a Bonafide record of the MAIN PROJECT work carried out by him under our guidance and supervision during the academic year 2025-26.

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EXTERNAL EXAM

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SHYAM KRISHNAN(MLM24MCA-2052)

ABSTRACT

The rapid modernization of educational frameworks often leads to the marginalization of regional cultural identities and localized environmental knowledge. Eco Legacy is an advanced, web-based Institutional Learning System (ILS) designed to bridge this gap by digitizing the rich tapestry of Kerala folklore and the ecological significance of the Western Ghats. Built using the Python Django framework, the platform moves beyond traditional static repositories by implementing a gamified learning environment. By incorporating mechanics such as Experience Points (XP), leveling systems, and "streaks," the platform fosters deep student engagement, transforming the study of heritage into an interactive, growth-oriented experience.

A core innovation of Eco Legacy is its integration of Artificial Intelligence for automated content curation. Utilizing pre-trained transformer models, the system enables educators to generate high-quality study materials and AI-driven quizzes instantly, ensuring that the curriculum remains dynamic and scalable. The user interface reflects this growth-centric philosophy through the Knowledge Bloom dashboard—a minimalist, modern UI designed to visually represent a student's academic evolution. By providing specialized roles for Admins, Teachers, and Students, the platform creates a cohesive ecosystem for the preservation of indigenous knowledge

Keywords: *Regional Heritage Education, Knowledge Bloom Visualization, Gamified Pedagogy, Regenerative Learning*

Mapping with Sustainable Development Goals (Mention the Goals)	SDG 4- Quality Education SDG 9- Industry, Innovation, and Infrastructure SDG11- Sustainable Cities and Communities

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LIST OF ABBREVIATIONS

ABBREVIATION	FULL FORM
AQG	AI Quiz Generator
CMS	Content Management System
LMS	Learning management
FLAN	Fined-tuned Language Net
T5	Text-to-Text Transfer Transformer
HF	Hugging Face